Shadow Amalgamation for the Patchwork-Bag Abstraction: a Round-13 Repair of Theorem B for $\mathcal{ALCI}_{\mathrm{RCC5}}$

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Abstract

The seventh review of the $\mathcal{ALCI}_{\mathrm{RCC5}}$ decidability project found that the round-12 Truth Lemma rests on an unproved *Shadow Amalgamation* assumption: the global relation ρ_T is amalgamated by patchwork from the bag networks only, while the lemma silently assumes the declared relation-vector shadow entries agree with ρ_T; for the legality tests as written this is false (a machine-checked three-bag configuration passes every test yet admits no realizing frame). This manuscript repairs Theorem B by *tightening the abstraction language*: the **D-discipline** restricts shadow entries to the horizontal relations $\{\mathsf{DR}, \mathsf{PO}\}$ (vertical and equality relationships must be co-bagged, which is the existing saturation and residual-frontier regime extended to the shadow layer), makes obligation shadows propagate mandatorily with a verticalization fallback, propagates transitive vertical universals through types, keeps request sources live, and adds a pair-coherence rule. Under this discipline the **Shadow Amalgamation Lemma** is proved: bag networks, all declared rows, and a canonical completion of the remaining shadow pairs are jointly realizable in one abstract RCC5 frame. The proof is a short case analysis over four finite composition-table facts, followed by the round-12 patchwork-plus-compactness argument applied to the shadow-extended family; every finite ingredient is machine-checked (WP17), and the repaired truth lemma then goes through with the declared entries literally equal to ρ_T. The cost of the repair is made explicit: the residual proof burden moves to the *completeness* side (every saturated split-forest presentation must be abstractable within the discipline), which is discharged by a Coverage Lemma leaning on the round-12 saturation rules, and flagged as the remaining keystone together with the (repaired) effective width bound. Status: the decidability theorem remains *strongly supported, not certified.*

Contents

*Round-13 repair manuscript, authored by the *seventh reviewer* of this project (the hot audit of 2026-07-08, `papers/fable5_review/`); it repairs the defects that review found in GPT-5.5's round-12 manuscript. It is a *delta* on round-12: everything not amended here is inherited from `split_forest_automata_with_appendices.tex` (May 31, 2026). This manuscript has *not* been reviewed, cold or otherwise; the usual project convention applies (seven reviews so far, seven defects; do not describe the theorem as proven or certified).

1 What this manuscript is, and its honest standing

This is a *repair delta* on the round-12 manuscript [1]: it amends the abstraction language of Theorem B, proves the lemma whose absence the seventh review [2] identified as the critical gap (finding F1), and fixes the review's other findings (F2–F6). Everything not explicitly amended — the semantics, Theorem A (split-forest normal form), the patchwork theorem and its finite-to-countable passage (round-12 Theorem 2.5 and Appendix A), the two-way alternating parity automaton frame and its emptiness reference — is inherited unchanged.

Two disclosures. First, the author of this repair is the seventh reviewer; the manuscript is therefore *not* an independent check of its own claims, and by the project's own standard it must be cold-reviewed before its status can improve. Second, the repair deliberately *trades directions*: round-12's broken step was in the *realization* direction (a valid abstraction must unfold to a model); this manuscript proves that direction fully, *given* the tightened discipline, and in exchange the *extraction* direction (every saturated presentation must be expressible within the tightened discipline) carries a larger burden than before. Section 6 discharges it by a Coverage Lemma whose load-bearing inputs are exactly the round-12 saturation rules (S1)–(S6); the residual risk of this manuscript is concentrated there and in the width interaction of Section 8, and we say so plainly in Section 11.

Machine support: the finite composition facts, the constructive assignment, and the joint-realizability claim of the central lemma are verified by `verification/python/wp17_round13_discipline_check.py` (400/400 random tree-shaped configurations under the discipline; negative control 151/300 failures when the discipline is violated); the configuration-level defect being repaired, and the survival of the discipline against it, were established by WP15; the acceptance-level behaviour of exactly this discipline is the "repaired mode" of WP16 (all ten referee diagnostics agree with the cover-tree tableau; a 148-concept random sweep with zero disagreements).

2 Inherited setting and notation

We use round-12's setting verbatim: abstract complete atomic RCC5 frames (Δ, ρ) with $\rho : \Delta^2 \to \mathsf{Rel}$, $\mathsf{Rel} = \{\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}$, satisfying identity (R1), converse (R2), and composition (R3) for the standard RCC5 table; concepts in NNF over $\mathrm{Cl}(C_0)$; Hintikka types $\mathrm{Typ}(C_0)$; patchwork bags $B = (P, N, \tau, \mu, W, U, \Gamma)$ over width $K(C_0)$; trees of bags with overlap maps, connected occurrence traces (running intersection), and the validity clauses (V1)–(V8); the RCC5 patchwork theorem (round-12 Theorem 2.5) with its finite tree-patching and countable compactness lemmas (round-12 Appendix A); and the two-way alternating parity tree automaton. Write $\mathsf{Hor} = \{\mathsf{DR}, \mathsf{PO}\}$ (*horizontal* relations) and call $\mathsf{PP}, \mathsf{PPI}, \mathsf{EQ}$ *vertical*.

An *occurrence* is an overlap-equivalence class of ports. An occurrence is *live* at bag t if it has a port manifestation in P_t. A *shadow* of an occurrence x at bag t (where x is not live) is the round-12 relation-vector record $(t_x, \vec{r}, \Omega)$; we call $\vec{r}$ the *row* of x at t and $\Omega \subseteq \mathsf{Rel} \times \mathrm{Cl}(C_0)$ its obligation set, induced by the universal formulas of $\mathrm{tp}(x)$.

2.1 The four finite facts

The repair rests on four properties of the composition table, each a finite check (machine-verified exhaustively, WP17 part A, against a table independently re-derived from set semantics).

Fact 2.1 (Horizontal absorption)**.** For all $a, b \in \mathsf{Hor}$: $\mathsf{Hor} \subseteq \mathrm{Comp}(a, b)$.

Fact 2.2 (Vertical transitivity)**.** $\mathrm{Comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$ and $\mathrm{Comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$.

Fact 2.3 (No horizontal dead ends)**.** For all $h \in \mathsf{Hor}$ and $r \in \{\mathsf{PP}, \mathsf{PPI}\}$: $\mathrm{Comp}(h, r) \cap \mathsf{Hor} \neq \varnothing$.

Fact 2.4 (EQ is never forced)**.** Every cell of the table containing EQ, other than $\mathrm{Comp}(\mathsf{EQ}, \mathsf{EQ})$, also contains PP, PPI and PO.

Fact 2.1 is the engine of the whole repair: *a triangle two of whose sides are horizontal admits every horizontal third side.* Horizontal values never constrain one another through the composition table; all genuine constraint flows through vertical structure, and the discipline confines vertical structure to co-bagged ports, where the complete bag networks and the patchwork theorem already govern it.

3 The D-discipline

The following amendments to round-12's Definitions 3.3/3.4 and Appendix Q define the round-13 abstraction language. Each clause is a finite local test (or, for D2, a transition-function obligation), so the alphabet remains effectively enumerable and the automaton finite.

Definition 3.1 (D-discipline)**.** A valid patchwork abstraction additionally satisfies:

(D1) **Horizontal rows.** Every shadow row entry lies in Hor. In particular no entry is EQ (this subsumes the seventh review's F2 fix), and no entry is PP or PPI: a pair whose relation is vertical must be *co-bagged* — represented by port manifestations in a common bag — exactly as the saturation rules (S2)/(S3)/(S4) and the residual-frontier clause (V7) already require for side contexts. A row must be *one-point closed* with its bag: the network on $P_t \cup \{x\}$ obtained by adjoining the row to N_t satisfies the composition table on every triple.

(D2) **Mandatory propagation with verticalization fallback.** If $\Omega_x \neq \varnothing$, then a shadow of x is present in *every* direction out of every bag where x is live or shadowed: crossing a boundary, the row entries on continued ports are inherited (rigidity) and entries for new ports are declared, subject to one-point closure. If closure would force an entry outside Hor (the true relation is vertical), the crossing is replaced by *verticalization*: a manifestation of x is inserted into the target bag as a port (this is rule (S3)/(S4) applied to the shadow layer), after which propagation resumes from the new manifestation. A shadow is never silently dropped (the seventh review's F3 fix).

(D3) **Type propagation of transitive vertical universals.** If $\forall\mathsf{PP}.D \in \tau(p)$ and $N_t(p,q) = \mathsf{PP}$ then $\{D,\ \forall\mathsf{PP}.D\} \subseteq \tau(q)$; dually for PPI. (Sound by Fact 2.2; this is the `compute_safe` rule of the project's implementations, now a bag-level legality test.) Consequently vertical universal obligations never need long-range shadows: they ride the types along the co-bagged vertical skeleton.

(D4) **Request-source liveness.** A vertical request (PP, D) or (PPI, D) keeps a manifestation of its source occurrence live (co-bagged) along the declared support branch until fulfillment, so fulfillment ("the recorded relation to the source is R and $D \in \tau(q)$") is a bag-network fact, never a shadow fact. (This pins down round-12's (V8), the seventh review's F6(iii).)

(D5) **Pair coherence.** The declared relation between two given occurrences is unique: if x's row at bag t has an entry for a manifestation of y, and y's row at bag t' has an entry for a manifestation of x, the two entries are converse to each other. (A two-way automaton checks this by carrying the value along the tree path between the two declaration sites.)

Remark 3.2. D5 is not cosmetic. The WP17 harness, before modelling it, produced exactly the round-12 failure pattern from two shadows co-located near each other's live manifestations with an undeclared mutual value; with D5 the pattern is excluded and the sweep is clean. D5 is the shadow-layer analogue of overlap agreement (O1)–(O5).

Remark 3.3 (Local checkability)**.** (D1) and (D3) are per-bag table checks; (D5) is a per-pair regular check; (D2) and (D4) are conditions on the transition function (the automaton *must* spawn the copies; it may not choose scopes). All state components range over the finite sets of round-12 Appendix I, so finiteness of the automaton is unaffected. Alternation absorbs the unbounded *instance* counts (one run copy per shadow instance, finite state space of profiles), exactly as in round-12.

4 The Shadow Amalgamation Lemma

Fix a valid D-abstraction T. Let V_T be its occurrences. Define the *declared data* Λ as the partial Rel-labelling of V_T^2 consisting of: (i) all bag network values (on co-bagged pairs); (ii) all shadow row entries (on shadow–port pairs), which by D5 assign a unique value per pair; (iii) for every remaining pair of occurrences (never co-bagged, neither covering the other by a row — necessarily both obligation-free by D2): the value DR.

Lemma 4.1 (Extended-bag closure)**.** *For every bag t, the network on $P_t \cup S_t$ (where S_t is the set of occurrences shadowed at t) induced by Λ is a complete atomic composition-closed RCC5 network.*

Proof. Completeness: bag values cover P_t^2; rows cover $S_t \times P_t$ (D1/D2); clauses (ii)/(iii) of Λ cover S_t^2. Atomicity: bag networks are atomic with EQ only on the diagonal; row entries and default values lie in Hor, and distinct occurrences never receive EQ (D1). Composition closure, by the number of shadow members in a triple:

- *No shadow*: the triple lies in N_t, closed by (N6).
- *One shadow* (x, p, q): this is precisely the one-point closure test of D1.
- *Two or three shadows*: every side incident to a shadow lies in Hor (D1 for shadow–port sides; (ii)/(iii) for shadow–shadow sides). In each of the three cyclic constraint positions the composed pair of sides is horizontal–horizontal, so by Fact 2.1 the third side — itself horizontal — is admitted. $\square$

Lemma 4.2 (Agreeing tree family)**.** *The family $\{P_t \cup S_t\}_{t \in T}$ with the Λ-induced networks is tree-shaped (every occurrence appears in a connected set of extended bags) and adjacent extended bags agree on their common occurrences.*

Proof. Traces of ports are connected by (C1)–(C3); scopes of shadows are connected by construction of D2 (propagation crosses adjacent boundaries; verticalization inserts a manifestation in the adjacent bag, preserving connectedness of the combined trace-plus-scope). Agreement: bag values agree by (O3); row entries agree on continued ports by rigidity; shadow–shadow values are single global values per pair by D5 and clause (iii). □

Theorem 4.3 (Shadow Amalgamation)**.** *There is an abstract complete atomic RCC5 frame ρ_T on V_T extending Λ: every bag value, every declared row entry, and every default value of clause (iii) is realized verbatim.*

Proof. Every finite subfamily of extended bags, restricted to finitely many occurrences, is a finite tree-shaped family of finite complete atomic composition-closed networks agreeing on overlaps (Lemmas 4.1 and 4.2; restriction preserves closure because (R1)–(R3) are universal conditions). By round-12's finite tree-patching lemma (from the two-bag patchwork property, Theorem 2.5) each such subfamily has a satisfiable completion. The countable passage is round-12 Appendix A verbatim, applied to the extended family: the closed cylinder sets $C_S = \{\lambda \in \mathsf{Rel}^{V_T^2} : \lambda \restriction S$ is *some* complete atomic closed extension of the extended bags of $S\}$ have the finite intersection property (a common finite super-subtree witnesses it), and any point of the intersection is a total labelling satisfying (R1)–(R3) on every triple, since every triple of occurrences lies in some finite connected subtree of extended bags. Atomicity and (R1) hold because every finite completion is atomic with EQ only on identical occurrences (Fact 2.4 guarantees non-EQ choices are never excluded, as in round-12 Appendix G).[1] □

Remark 4.4 (Why the round-12 gap cannot recur)**.** The seventh review's counterexample declares $x_1 \, \mathsf{PP} \, p$ and $x_2 \, \mathsf{PPI} \, p$ as *shadow entries*; both are illegal under D1. Its exhaustive variant needs at least one vertical entry in every failing assignment (all 23 unrealizable assignments of the minimal pattern contain one). WP17's part C reproduces this: re-admitting vertical entries makes 151 of 300 sampled configurations unrealizable, while under the discipline 400 of 400 tree-shaped configurations both pass Lemma 4.1 and are globally realizable.

5 The repaired Truth Lemma

Let T be a valid D-abstraction with a root port containing C_0, and let ρ_T be the frame of Theorem 4.3. Interpret concept names by the types, as in round-12.

Lemma 5.1 (Truth lemma, round-13 form)**.** *For every occurrence x and $C \in \mathrm{Cl}(C_0)$: $C \in \mathrm{tp}(x) \iff \mathcal{I}_T, x \models C$.*

Proof. Induction on C; atomic and Boolean cases as in round-12.

Existentials. If $\exists R.D \in \mathrm{tp}(x)$ with a local (or saturated-side-context) witness, the witness edge is a bag value, so ρ_T realizes it verbatim (Theorem 4.3) and induction applies. If it is a vertical request, D4 keeps a manifestation of x co-bagged along the support branch; parity acceptance yields a bag in which the fulfilling port q satisfies $N(x, q) = R$, $D \in \tau(q)$ — again a bag value. Conversely if $\exists R.D \notin \mathrm{tp}(x)$, then $\forall R.\overline{D} \in \mathrm{tp}(x)$ and the universal case applies.

[1] The wording fix to round-12's Appendix A demanded by the seventh review's F5 ("*the* patchwork completion" → "*some* completion") is incorporated in the definition of C_S above.

Universals. Let $\forall R.D \in \mathrm{tp}(x)$ (so $\Omega_x \neq \varnothing$) and let y be any occurrence with $\rho_T(x, y) = R$. We claim $D \in \mathrm{tp}(y)$, whence induction finishes. Distinguish how the pair (x, y) is represented:

(i) *Co-bagged*: $\rho_T(x, y) = N_t(x, y)$ for a common bag t (Theorem 4.3 extends bag values), and the in-bag universal test (round-12 (T4), retained) forces $D \in \tau(y)$.

(ii) *Row-covered*: some row of x (or of y, via D5 the value is the same) declares the pair's value $e \in \mathsf{Hor}$. By Theorem 4.3, $\rho_T(x, y) = e$; the shadow obligation check fires on the entry $e = R$ and forces $D \in \tau(y)$. *This is the step that was unproved in round-12; it is now literal, because ρ_T extends the rows by construction.*

(iii) *Neither*: by the Coverage Lemma below (Lemma 6.1), for an obligation-carrying x this happens only when $R \in \{\mathsf{PP}, \mathsf{PPI}\}$ and y lies on the co-bagged vertical skeleton reachable from x by a chain of R-labelled bag edges; then D3 has propagated $\{D, \forall R.D\}$ along the chain into $\mathrm{tp}(y)$'s predecessors and D into $\mathrm{tp}(y)$ (soundness of the propagation is Fact 2.2: the composed relation along the chain is exactly R). For the default pairs of clause (iii), $\rho_T(x, y) = \mathsf{DR}$ with both sides obligation-free, so no universal at x exists to check — excluded here since $\Omega_x \neq \varnothing$ means x is not default-paired (D2).

The converse direction ($\forall R.D \notin \mathrm{tp}(x)$ gives a counter-witness) is by closure under NNF complement and the existential case, as in round-12. □

6 The Coverage Lemma (the completeness side)

The truth lemma used:

Lemma 6.1 (Coverage)**.** *In a valid D-abstraction, for every occurrence x with $\Omega_x \neq \varnothing$ and every occurrence y, at least one of: (a) x, y are co-bagged; (b) the pair's value is declared by a row (of x or of y); (c) $\rho_T(x, y) \in \{\mathsf{PP}, \mathsf{PPI}\}$ is realized along a co-bagged chain of vertical bag edges from x to y (through which D3 propagates).*

Proof. Let d be the tree distance between the sets of bags carrying x (live or shadow) and those carrying y. If $d = 0$ they share a bag: if both live, (a); if one is shadowed there, its row covers the other's manifestation (rows cover all ports, D2), giving (b). For $d > 0$, D2 propagates x's shadow along the connecting path. At each boundary the crossing either succeeds with horizontal entries — decreasing d — or triggers verticalization, which inserts a manifestation of x into the next bag as a *port*, also decreasing d; by Fact 2.3 these are the only cases (a horizontal anchor is never composition-forced into "no horizontal and no verticalization" — if every closure-consistent entry is vertical, verticalization applies; if none exists at all, the abstraction is illegal and rejected). By induction on d, either x's manifestations reach a bag of y (cases (a)/(b)), or the walk consists entirely of verticalization steps whose inserted manifestations are related to their predecessors by bag edges labelled PP (resp. PPI) throughout; in that case the composed relation is exactly PP (resp. PPI) by Fact 2.2, giving (c). Mixed walks (some horizontal crossings, then verticalizations) end in cases (a)/(b) at the last horizontal manifestation's bag family. □

Remark 6.2 (Where the burden sits)**.** Lemma 6.1 is a statement about *valid D-abstractions.* The companion obligation — that every satisfiable concept *has* a valid D-abstraction — is the extraction direction, Section 7: there, the model supplies the true relations, and the discipline's demands (horizontal rows, verticalization, liveness) are met exactly by the saturation apparatus (S1)–(S6) of Theorem A. This inheritance is deliberate and is flagged in Section 11: Theorem A's saturation has been found sound by every review, but the *interaction* of on-demand verticalization with the width bound is the part of this repair most deserving of the next adversarial review.

7 Extraction under the discipline (Theorem B(a) delta)

Let Φ be a saturated split-forest presentation of width $K(C_0)$ (Theorem A). Round-12's extraction (its Appendix D/M) is amended:

(E1') Rows are the *true* model relations from forgotten sources to current ports. Whenever the true relation is vertical or EQ, the presentation does not shadow it: saturation (S3) verticalizes the pair (inserting the stratum or overlap identity), (S4) splices infinite tails at bounded thresholds, and (S2) keeps the relevant vertical boundary co-bagged. Hence all row entries are horizontal: D1 holds.

(E2') Obligation sources' shadows are recorded in every direction; where the true relation to a boundary turns vertical, the (S3)/(S4) step of (E1') has already co-bagged the pair, so the fallback of D2 is exactly the saturation step: D2 holds.

(E3') Types are read off the model; since the model satisfies $\forall\mathsf{PP}.D$ at p and $\rho(p,q) = \mathsf{PP}$ implies both D and (by Fact 2.2) $\forall\mathsf{PP}.D$ hold at q, the D3 clauses are true type memberships: D3 holds.

(E4') Vertical requests name their true witnesses; the source is kept live along the finite support prefix (its trace is extended — a presentation choice, not a semantic change): D4 holds.

(E5') Entries are true relations, so the value of a pair is unique: D5 holds.

Since all declared data are restrictions of one real model, all local tests pass, and the extraction lands in the (still finite) alphabet.

8 The effective width bound, repaired (F4)

Round-12's Lemma 3.5 asserted finiteness of the saturated-context classes by an enumeration that is a priori infinite (the seventh review's F4; the sixth review's M1). We replace it.

Definition 8.1 (Witness-generated context)**.** A saturated side context is *witness-generated* if every port carries a *generation reason*: (g1) the source u itself; (g2) a selected witness of an existential formula of $\mathrm{Cl}(C_0)$ at a port already generated; (g3) a boundary insertion under (S2) (live parent, live children, support endpoints, overlap manifestations of an already generated port); (g4) a vertical stratum or threshold node inserted by (S3)/(S4) for a pair of already generated ports; (g5) a port of a recursive side subcontext (rank strictly smaller).

Lemma 8.2 (Effective bound)**.** *Let $n = |\,\mathrm{Cl}(C_0)|$. There are computable numbers $c(n)$ and B_r such that every rank-r witness-generated saturated context has at most B_r ports, where $B_0 \leq c(n)$ and $B_{r+1} \leq c(n)\,(1 + B_r)^2$. Hence $K(C_0) \leq B_n$ is computable.*

Proof. Reasons (g2) contribute at most n ports per generated port (one per existential formula of the closure); (g3) at most $2 + n + 2$ per generated port (parent, children bounded by the number of existentials, at most two support endpoints, plus overlap manifestations bounded by the number of incomparable superpart witnesses $\leq n$) — absorb all these constants into $c(n)$. (g4) inserts at most one port per *pair* of generated ports, contributing quadratically; (g5) attaches, for each of the at most n recursive triggers per generated port, a context of rank $< r$, of size $\leq B_{r-1}$ by induction. Every port has a finite generation chain terminating in (g1), so the whole context is contained in the union counted by the recurrence; ranks are bounded by n because recursive triggers descend to proper subformulas. All quantities are computable. □

Only witness-generated contexts are enumerated (extraction produces no others: every inserted port in Section 7 has one of the reasons (g1)–(g5)); the enumeration of round-12's Appendix K is run with the size cap B_r per rank, and its termination is now a theorem, not an observation. No optimality is claimed.

9 Automaton delta and remaining formalities (F5, F6)

(a) *Transitions.* Shadow states additionally reject EQ entries and non-horizontal entries (D1); obligation shadows are spawned universally into all directions (D2) — alternation makes this a conjunction in the transition formula, not a choice; verticalization is a labelled alternative checked against the bag (D2); the D3 clauses join the local type tests; request states carry the source port through overlaps (D4); D5 is checked by a two-way excursion between the two declaration sites of a pair, carrying one Rel value.

(b) *Parity.* Each request copy carries an odd priority until it enters its fulfilled state (even); infinite deferral of any single request is thus rejecting. No inter-request round-robin is needed under alternation. (This replaces round-12's generalized-Büchi narrative; the seventh review's F6(i), and its incidental finding that naive same-profile blocking without request-closed laps immediately accepts a false SAT, WP16.)

(c) *Input trees.* Trees are padded to a fixed maximal branching degree computable from $K(C_0)$ and n, with a dummy label (F6(ii)).

(d) *Appendix A wording.* Incorporated in Theorem 4.3's proof (F5).

The state components remain within round-12's Appendix I bounds (profiles, request kinds, pair-check values), so the automaton is finite and emptiness is decidable (Vardi).

10 Diagnostics and machine checks

The acceptance behaviour of exactly this discipline is WP16's repaired mode: agreement with the cover-tree tableau on all ten referee diagnostics — C_{force}, C_{split}, C_{2hop}, C_{3hop}, $\exists\mathsf{DR}.A \sqcap \forall\mathsf{DR}.\neg A$ unsatisfiable; $C_{\text{recursive}}$, $C_{\uparrow}$, the dual-descendant tower, the strong-EQ PO-loop, and a sanity concept satisfiable — plus a 148-concept random sweep with zero disagreements. The PO-loop case is worth singling out: exact occurrence identity with same-bag witness reuse decides it correctly by design, where the project's retired quasimodel oracle failed. WP17 verifies Facts 2.1–2.4 exhaustively and the constructive content of Lemmas 4.1–4.3 on 400 random tree-shaped configurations (all pass), with a functioning negative control (151/300 failures when D1 is dropped). WP14 backs Theorem 2.5 itself; WP15 delimits the defect this manuscript repairs.

11 What would invalidate this repair

(G1) A valid D-abstraction whose declared data Λ is not jointly realizable would refute Theorem 4.3; by Lemma 4.1 it would have to violate one of the four finite facts or the family structure (WP17 searched; none found).

(G2) A satisfiable concept none of whose saturated presentations can be expressed within the D-discipline would break extraction (Section 7). The pressure point is on-demand verticalization versus the width bound: if keeping obligation sources co-bagged through vertical regions (D2 fallback, D4 liveness) could force unboundedly many simultaneous manifestations *not* absorbed by the witness-generated accounting of Lemma 8.2, the alphabet would no longer cover all models. This is the direct descendant of the round-9/10 forced-verticalization problems (machine-checked at the certificate level by WP10–WP12) and is the **designated next review target**.

(G3) An error in the Coverage Lemma's walk argument (Lemma 6.1), e.g. a pair reachable only by a walk that alternates horizontal and vertical crossings in a way that terminates in neither (a)/(b) nor a pure vertical chain (c).

(G4) Failure of either inherited external theorem (RCC5 patchwork; two-way alternating parity emptiness).

12 Status

With this repair: the *realization* direction of Theorem B — where rounds 11 and 12 broke — is proved from four machine-checked finite facts plus the patchwork/compactness apparatus that WP14 verifies in exactly the form used. The *extraction* direction and the width accounting now carry the residual risk, localized in Section 11(G2)–(G3). The decidability theorem for $\mathcal{ALCI}_{\mathrm{RCC5}}$ over abstract complete atomic RCC5 frames therefore remains, in the project's calibrated language, **strongly supported, not certified**: this manuscript is unreviewed, was written by the seventh reviewer, and by the project's own track record (seven reviews, seven defects) should be presumed to contain a defect until a genuinely cold review fails to find one.

References

[1] Consolidated proof manuscript (GPT-5.5, OpenAI). *Decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ via Saturated Split Forests, Patchwork Bags, and Two-Way Parity Automata*, round-12 expanded self-contained edition, May 31, 2026. `papers/automata_route_repairs/split_forest_automata_with_appendices.tex`.

[2] Claude (Fable 5, Anthropic). *Referee Report and Computational Stress Test: the Round-12 Patchwork-Bag Automata Proof of Decidability for $\mathcal{ALCI}_{\mathrm{RCC5}}$*, July 8, 2026. `papers/fable5_review/referee_report_round12_fable5.pdf`.

[3] J. Renz and B. Nebel. On the complexity of qualitative spatial reasoning. *Artificial Intelligence* 108 (1999).

[4] C. Lutz and M. Miličić. A tableau algorithm for description logics with concrete domains and GCIs. *J. Automated Reasoning* 38 (2007).

[5] M. Y. Vardi. Reasoning about the past with two-way automata. *ICALP* 1998.